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Abstract

Student academic performance is affected by factors such as study habits, motivation, sleep, parental involvement, and technology use. This study uses statistical analysis and multiple linear regression to examine how these factors influence exam scores using a Kaggle dataset of 6,607 students. Pearson correlation, *t*-tests, and ANOVA were applied to identify significant relationships. The results show that study hours, motivation, parental involvement, and internet access significantly affect academic performance, with study hours being the strongest predictor. The regression model explains about 21% of the variation in exam scores, providing useful insights for data-driven educational support.

1.0 Introduction

Academic performance is influenced by both internal factors, such as study habits and motivation, and external factors, including parental involvement and technology use. With the increasing use of digital tools, understanding the impact of lifestyle and technology on learning has become important. However, many educational interventions lack a clear, data-driven basis. This study applies statistical testing and regression analysis to identify key factors affecting students' exam performance and support more effective educational decision-making.

1.1 Background of study

1. Factors Influencing Student Academic Performance

Understanding these factors is important to improving student achievement and identifying the most relevant factors influencing students' academic success. Past research has shown that students' grades are significantly affected by internal factors like motivation, study habits, and sleep duration.[1] In further, performance is also impacted by outside factors, such as family support, the environment of school, and availability of educational resources. [2] In addition, social and psychological factors like emotional stability and physical exercise influence how well students perform in their academics. [3]

The purpose of this study is to find out how these factors, especially study habits, the learning environment, motivation, and parental support affect the student's academic performance. The research aims to provide information that can help educators as well as organizations build specific approaches to improve student learning outcomes by identifying the most important factors.

2. Lifestyle and Technological Impact on Student Performances.

The use of technology and daily habits have an important effect on student academic performance. Concentration, motivation, and learning ability are all directly affected through variables including screen time, physical exercise, and sleep quality. Students who consistently and properly sleep perform better academically, underlining the significance of maintaining healthy daily habits. [4]

Students' ability to focus and organize their study time is also being affected by the increasing usage of digital devices. While technology makes educational resources more accessible, too much screen time may lead to distractions and lower learning effectiveness.[5] A study found a negative correlation between kids' academic performance and extended screen time, showing that digital balance is important factors for affecting educational results.

1.2 Research Objective

1. Identify and classify the key internal and external factors affecting student academic performance using a Kaggle open source dataset.
2. Analyze the statistical relationships between these factors and academic results using appropriate statistical tests such as Chi-square, ANOVA, and t-tests.
3. Develop a multiple linear regression model to predict student performance based on significant variables, targeting at least 80% accuracy in model validation.
4. Evaluate the model's reliability using metrics such as R^2 , Adjusted R^2 , and RMSE.
5. To examine the impact of technology usage, including screen time and digital learning engagement, on students' academic outcomes through regression analysis and data visualization.

2.0 Problem Statement

1. Lack of a Data-Driven Model for Predicting Student Academic Performance

Study habits, motivation, sleep quality, parental support, and technology use are just a few of the many internal and external factors that interact to influence students' academic performance. However, there is currently a lack of a comprehensive, data-driven statistical model that measures how each of these measurable variables affects students' academic performance. Teachers and organizations are unable to identify the elements that have the biggest influence on academic achievement in the lack of such an independent methodology. Predictive analytics using educational data may significantly enhance organizational assessment, but many present techniques are unable to effectively incorporate a variety of behavioral and lifestyle factors.[6]

2. Ineffective and Non-Targeted Educational Interventions

Intervention programs are often implemented through educational organizations without a clear understanding of the particular characteristics that have the biggest effect on student performance. Because of this, these strategies usually fail to meet the various educational requirements of students and are ineffective. Program effectiveness is limited, vulnerable kids are not identified promptly, and resources are allocated poorly due to a lack of statistical information into academic performance factors. In order to promote effective strategies, schools can identify crucial variables that impact learning outcomes through the development of an accessible prediction model. Highlights how machine learning and regression models are used in education to optimize resource allocation, customize learning support, and increase organizational efficiency.[7]

3. Unclear Impact of Technology Use on Academic Performance

Technology use has become a significant factor impacting academic achievement due to the quick incorporation of digital devices, social media, and Artificial intelligence

(AI) learning into students' daily activities. However, its precise effect remains unclear and underexplored in quantitative educational models. While limited, purposeful use of technology could enhance learning results, excessive screen time may have negative impacts on motivation, focus, and sleep. Teachers are unsure of how they can maintain a balance between digital involvement and academic success in the lack of data-driven insights. In order to effectively analyze the complicated and multifaceted relationship between screen media use and academic outcomes, data-based techniques are necessary.[8]

3.0 Literature Review

3.1 Impact of Study Habits on Academic Performance

Students' academic performance is significantly influenced by their study habits. Exam performance is much improved by organized study schedules, effective time management, and regular study hours [1]. Their results also highlighted how important sleep quality is, since obtaining sufficient sleep improves mental agility, retention of information and overall educational effectiveness. The results show a high relationship between university students' improved academic performance and healthy sleep habits [4]. Each of these research supports the concept that getting enough rest and maintaining a structured study style are important internal components of academic achievement.

3.2 Role of External and Environmental Factors

Academic achievement is also influenced through external factors like family background, learning environments, and access to educational resources. Student achievement is mostly affected by the learning environment, which includes the classroom environment, the availability of study resources, and instructor support [2]. Additionally, socioeconomic and environmental factors have significant effects on student academic performance, motivation, and attention [3]. These results show that students' learning environments have an impact on their academic performance in addition to their own behaviors.

3.3 Lifestyle and Physical Activity Effects

Engaging in physical activity enhances persistence, emotional stability, and memory retention, all of which lead to better academic results [3]. Regular physical activity enhances students' ability to learn by lowering stress levels and increasing mental clarity. These findings show that holistic well-being is just as important to academic success as mental engagement.

3.4 Technology and Screen Time Effects on Academic Outcomes

Academic performance has both opportunities and problems as digital technology becomes increasingly incorporated into students' daily lives. Digital tools can encourage learning by

making material more accessible, but excessive or uncontrolled screen time can have a negative impact on motivation, focus, and sleep quality. Long-term screen media use is negatively correlated with children's and teenagers' academic performance, underscoring the possibility of distraction and decreased academic effectiveness [5]. This negative correlation was confirmed when screen time was unconnected to educational activities [8]. These results highlight how crucial it is to apply technology in balance to promote learning rather than obstruct it.

4.0 Dataset Selection & Justification

4.1 Dataset Source and Key Features

```
'data.frame': 6607 obs. of 11 variables:
 $ Hours_Studied      : int  23 19 24 29 19 19 29 25 17 23 ...
 $ Attendance         : int  84 64 98 89 92 88 84 78 94 98 ...
 $ Parental_Involvement: chr  "Low" "Low" "Medium" "Low" ...
 $ Access_to_Resources : chr  "High" "Medium" "Medium" "Medium" ...
 $ Sleep_Hours        : int  7 8 7 8 6 8 7 6 6 8 ...
 $ Motivation_Level    : chr  "Low" "Low" "Medium" "Medium" ...
 $ Internet_Access     : chr  "Yes" "Yes" "Yes" "Yes" ...
 $ Tutoring_Sessions   : int  0 2 2 1 3 3 1 1 0 0 ...
 $ Peer_Influence      : chr  "Positive" "Negative" "Neutral" "Negative" ...
 $ Physical_Activity    : int  3 4 4 4 4 3 2 2 1 5 ...
 $ Exam_Score         : int  67 61 74 71 70 71 67 66 69 72 ...
```

Figure 4.1.1 Data structure

The data that we have selected to conduct our research is the one entitled Student Performance Factors and obtained from Kaggle. This data has 6607 records of students with 20 variables, 19 of which are independent variables and the other 1 dependent variable. It has both the categorical, ordinal and the continuous variables that encompass a broad spectrum of academic and behavioural variables that might affect the performance of a student.[9]

This research includes a dependent variable, ExamScore, is the academic performance of students. The selected 10 independent variables from the 19 available variables in the dataset, focusing on those most relevant to factors influencing students' academic performance which include:

Table 4.1.2 Data Description for Variables

Variable	Description	Data Type	Measurement Scale
Study Hours	Average number of study hours per day.	Quantitative (Continuous)	Ratio
Parental Involvement	Level of family engagement in academic activities.	Quantitative (Categorical)	Ordinal
Sleep Hours	Average number of sleep hours per day.	Quantitative (Continuous)	Ratio
Level of Motivation	Self-rated motivation level to study.	Quantitative (Discrete)	Ordinal
Internet Usage	Average daily time spent on screens and online study activities.	Quantitative (Continuous)	Ratio
Exam Score (Dependent)	The average performance outcome in examinations.	Quantitative (Continuous)	Ratio

The dataset combines internal (motivation, study habits, sleep quality) and the external (parental support, learning environment, technology usage) variables, which makes it very relevant in our case. The diversity makes it possible to perform a more detailed statistical testing and predictive models development with the help of such tools as ANOVA, Chi-square, and multiple linear regression.

4.2 Alignment with Research Objectives

The data set contains both internal (motivation, sleep, study habits) and external (parental support, technology use) variables.

- Analyse statistical relationships using Chi-square, the analysis of variance (ANOVA) and the t-tests. Presence of categorical and continuous variables enables the means of groups to be compared and tests of independence conducted.
- Develop a multiple linear regression model used to predict academic performance. ExamScore is the dependent variable that has multiple independent variables that can be applied using regression.

- Evaluate the reliability of the model by using Rsquared, Adjusted Rsquared, and Root Mean Squared Error. Continuous data can be used to assess the model performance with the help of regression metrics.
- To examine the effect of the use of technology on academic achievement. The variable InternetUsage represents the relationship between digital engagement and its relationship with academic performance.

4.3 Challenges associated with the dataset

The data set is mostly well structured, but there are some pre processing steps that may be required:

- **Handling Missing Values:** Some of the records might contain incomplete information for some variables, such as SleepHours or ParentalInvolvement. Missing values will be imputed (based on the mean, mode, or by using regression-based imputation).
- **Encoding of Categorical Variables:** Qualitative information (MotivationLevel, ParentalInvolvement, etc.) will need to use label encoding or one-hot encoding to use in statistical and regression models.
- **Normalisation and Standardisation:** Continuous variables (such as study hours, sleep or exam score) will be scaled to give the same weight in analysing results.
- **Outlier Detection:** Outliers in numeric variables (e.g. extreme study hours) will be detected using the IQR or Z-score approach to avoid distorting the coefficients of the model.
- **Multicollinearity Assessment:** Variance Inflation Factor (VIF) tests will be used for detecting the potential overrelation between variables that may impact the accuracy of the regression process.
- **Data Splitting to Validation of the Created Model:** The dataset will be split into 80% training and 20% testing datasets in order to test model performance and prevent overfitting.

4.4 Data Summary and Cleaning

The purpose of this summary is to understand the distribution of data, identify missing values, and gain initial insights into the factors that may influence students' exam performance.

1. Data Summary

```
> summary(df)
Hours_Studied      Attendance      Parental_Involvement  Access_to_Resources  Sleep_Hours
Min.   : 1.00      Min.   : 60.00      Length:6607          Length:6607          Min.   : 4.000
1st Qu.:16.00      1st Qu.: 70.00      Class :character      Class :character      1st Qu.: 6.000
Median :20.00      Median : 80.00      Mode  :character      Mode  :character      Median : 7.000
Mean   :19.98      Mean   : 79.98                      Mean   : 7.029
3rd Qu.:24.00      3rd Qu.: 90.00                      3rd Qu.: 8.000
Max.   :44.00      Max.   :100.00                      Max.   :10.000
```

Figure 4.4.1 Data Summary

```
Motivation_Level  Internet_Access  Tutoring_Sessions  Peer_Influence  Physical_Activity  Exam_Score
Length:6607      Length:6607      Min. :0.000      Length:6607      Min. :0.000      Min. : 55.00
Class :character  Class :character  1st Qu.:1.000    Class :character  1st Qu.:2.000    1st Qu.: 65.00
Mode :character  Mode :character  Median :1.000    Mode :character  Median :3.000    Median : 67.00
                  Mean :1.494                      Mean :2.968      Mean : 67.24
                  3rd Qu.:2.000                      3rd Qu.:4.000    3rd Qu.: 69.00
                  Max. :8.000                      Max. :6.000      Max. :101.00
```

Figure 3.4.2 Data Summary

2. Missing Value

```
> colSums(is.na(df))
      Hours_Studied      Attendance      Parental_Involvement      Access_to_Resources      Sleep_Hours
               0               0               0               0               0
> |
```

Figure 3.4.3 Missing Value

```
Motivation_Level  Internet_Access  Tutoring_Sessions  Peer_Influence  Physical_Activity  Exam_Score
               0               0               0               0               0               0
```

Figure 3.4.4 Missing Value

There is no missing data in this dataset for all the variables. This shows that the data in our datasets are very clean without any missing data that potentially impact the validity, reliability, and outcomes of our analysis.

5.0 Hypothesis Formulation

This section lists the hypotheses based on the *Student Academic Performance Factors* dataset.

Each hypothesis shows how certain factors (independent variables) may affect students' exam scores (dependent variable).

These relationships will be tested using statistical analysis and multiple linear regression.

1. Hypothesis 1 – Study Habits and Exam Performance

Null Hypothesis (H_{01}):

There is no significant relationship between students' daily study hours and their exam scores ($\beta_1 = 0$).

Alternative Hypothesis (H_{11}):

There is a positive relationship between study hours and exam scores ($\beta_1 > 0$).

Reason:

Students who spend more time studying usually understand the material better and

perform higher in exams. This checks if more study time truly helps improve exam results.

2. Hypothesis 2 – Internet Usage and Exam Performance

Null Hypothesis (H_{02}):

Internet usage has no significant effect on students' exam scores ($\beta_2 = 0$).

Alternative Hypothesis (H_{12}):

There is a negative relationship between internet usage and exam scores ($\beta_2 < 0$).

Reason:

While the internet can support learning, using it too much can cause distractions. This tests whether spending a lot of time online lowers academic performance.

3. Hypothesis 3 – Parental Involvement and Exam Performance

Null Hypothesis (H_{03}):

Parental involvement does not affect students' exam scores ($\beta_3 = 0$).

Alternative Hypothesis (H_{13}):

Parental involvement has a positive effect on exam scores ($\beta_3 > 0$).

Reason:

Students who get guidance and support from their parents often stay more focused and motivated. This checks if students with supportive parents achieve better results.

4. Hypothesis 4 – Combined Factors and Exam Performance

Null Hypothesis (H_{04}):

The combination of study hours, sleep hours, motivation level, and internet usage does not significantly predict exam scores ($\beta_i = 0$ for all i).

Alternative Hypothesis (H_{14}):

At least one of these factors—study hours, sleep hours, motivation level, or internet usage—has a significant effect on exam scores, and the model explains part of the variation ($R^2 > 0$).

Reason:

This checks how these factors together affect exam results. If the model shows significance, it means these factors as a group can help explain students' exam performance.

6.0 Statistics Testing

This chapter is to determine whether the relationships suggested in the research hypotheses are validated by statistical testing. These tests help to identify whether elements like study time, internet usage, parental involvement, and motivation level have a major impact on student exam performance. The appropriate statistical methods are applied based on the types of variables which involve, pearson correlation and regression coefficient test are used to examine the impact of study habits, independent samples t-test evaluates differences in exam scores between high and low internet usage groups and ANOVA tests whether parental involvement levels produce meaningful variation in performance. Through these analyses help to validates the hypotheses and identifies which factors meaningfully contribute to academic outcomes, forming the empirical foundation for the study's conclusions.

6.1 Statistics Testing 1: Pearson Correlation Testing

The purpose of the Pearson correlation test is to determine the direction and strength of a linear relationship between two continuous numerical variables. Both the exam score (percent or points) and the number of hours studied (hours) are listed as numerical values in the dataset. Both variables match the basic prerequisite to implementing Pearson's test since they are continuous and neither categorized nor ordinal. Because of this, Pearson is the most simple and suitable approach for analyzing the relationship between these two quantitative variables.

Pearson correlation objective is to identify:

- Whether a relationship exists
- Whether it is positive
- How strong it is
- Whether the result is statistically significant

6.1.1 Data Visualization 1: Scatter plot of Hypothesis 1

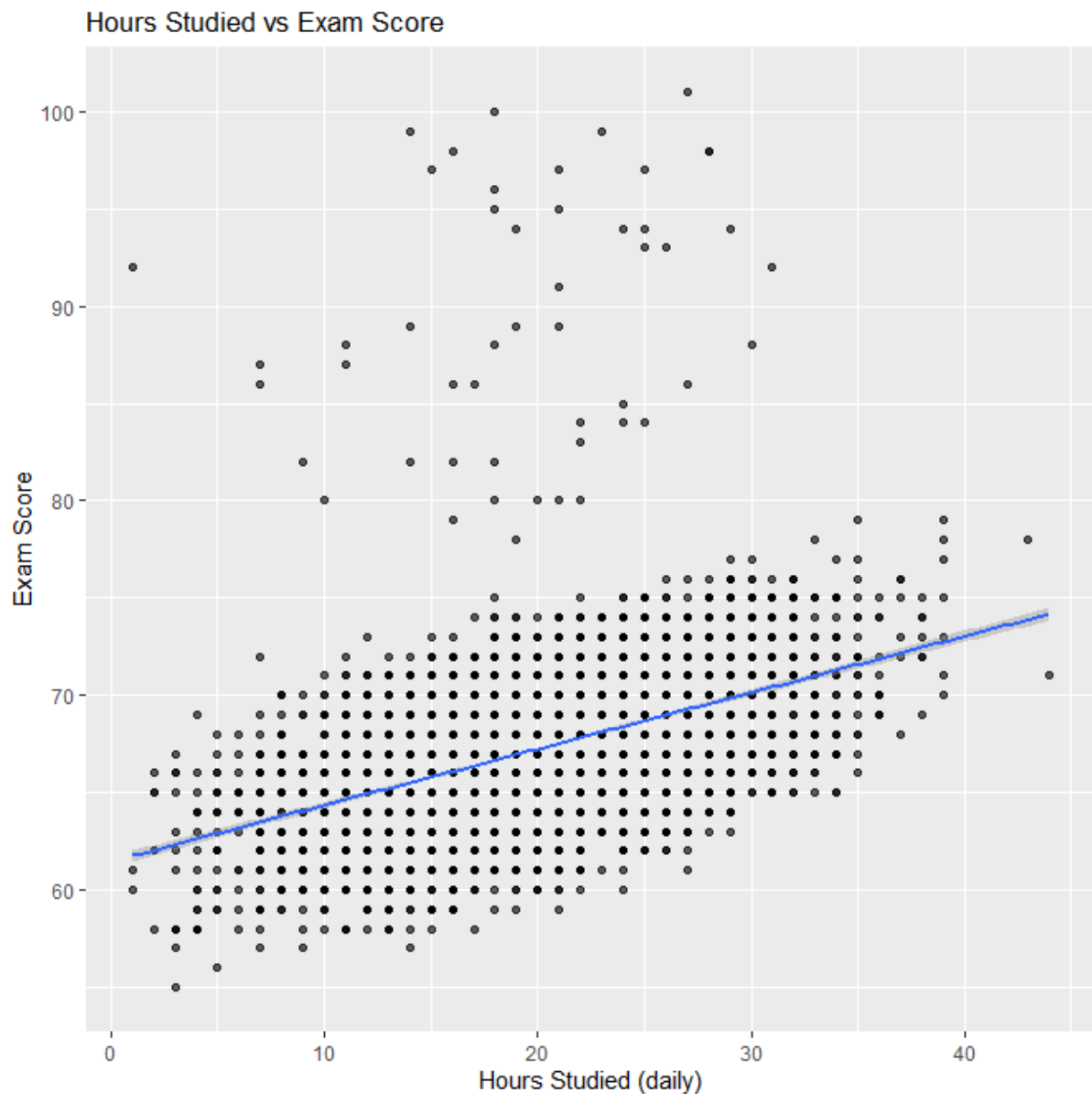


figure 6.1.1.1 Scatter plot of Hours Studied vs Exam Score

The scatter plot shows the relationship between Hours Studied (daily) and Exam Score. Each dot represents an individual student's data point. The overall pattern of the dots indicates a positive linear relationship as students spend more hours studying per day, their exam scores tend to increase.

A regression line which is the blue line included to highlight the trend. The upward slope of this line suggests that higher study hours are generally associated with higher exam performance. Although the data points show some variability meaning exam scores do not increase perfectly with study hours the overall direction still supports Hypothesis 1 (students who study more tend to achieve better exam results).

This visualization is clear, well-labeled, and professional, allowing readers to easily understand both the distribution of data and the trend being investigated.

6.1.2 Result and Conclusion

```
> cor_test_h1 <- cor.test(df$hours_studied, df$exam_score, method = "pearson", alternative = "greater")
> cor_test_h1

Pearson's product-moment correlation

data: df$hours_studied and df$exam_score
t = 40.436, df = 6605, p-value < 2.2e-16
alternative hypothesis: true correlation is greater than 0
95 percent confidence interval:
 0.4290853 1.0000000
sample estimates:
      cor 
0.445455
```

figure 6.1.2.1 Pearson Correlation Testing of Hypothesis 1

Pearson correlation results:

- The Pearson correlation test showed a moderate positive correlation between hours studied and exam score ($r = 0.445$).
- The correlation was statistically significant with an extremely small p-value ($p < 2.2e-16$), indicating the result is not due to random chance.
- The one-tailed test confirmed the correlation is significantly greater than zero, supporting the hypothesis that studying more is associated with higher exam performance.
- The 95% confidence interval suggested the true correlation is at least 0.429, meaning the positive association is consistently present in the population.
- Overall, the findings strongly support the alternative hypothesis that increasing study hours is linked to improved exam scores.

Formula:

$$r = \frac{\sum(x - m_x)(y - m_y)}{\sqrt{\sum(x - m_x)^2 \sum(y - m_y)^2}}$$

Where:

- **r** : pearson correlation coefficient
- **x and y** : two vectors of length n
- m_x and m_y : corresponds to the means of x and y.

Conclusion:

The Pearson correlation test is appropriate because both variables are continuous, the study aims to test a linear relationship, and the dataset is large enough to satisfy test assumptions. The output shows a moderate positive correlation ($r = 0.445$) with an extremely small p-value ($< 2.2e-16$), indicating strong evidence that increased study hours are significantly associated with higher exam scores.

6.2 Statistics Testing 2: Two-sample t-test

The study analyzes the mean exam scores of two independent groups which are students with and without internet access by using a continuous outcome variable, which makes the two-sample t-test acceptable. The test focuses on the study problem since the groups are made up of various people and the objective is to see whether their average scores range. Two-sample t-test is especially suitable because the group sizes are unequal and equal variances cannot be assumed. The two-sample t-test is a reliable and valid technique to determine the mean difference between the two groups since the test's assumptions are well-supported by the large sample sizes in both groups.

6.2.1 Data Visualization 2: Boxplot of Hypothesis 2

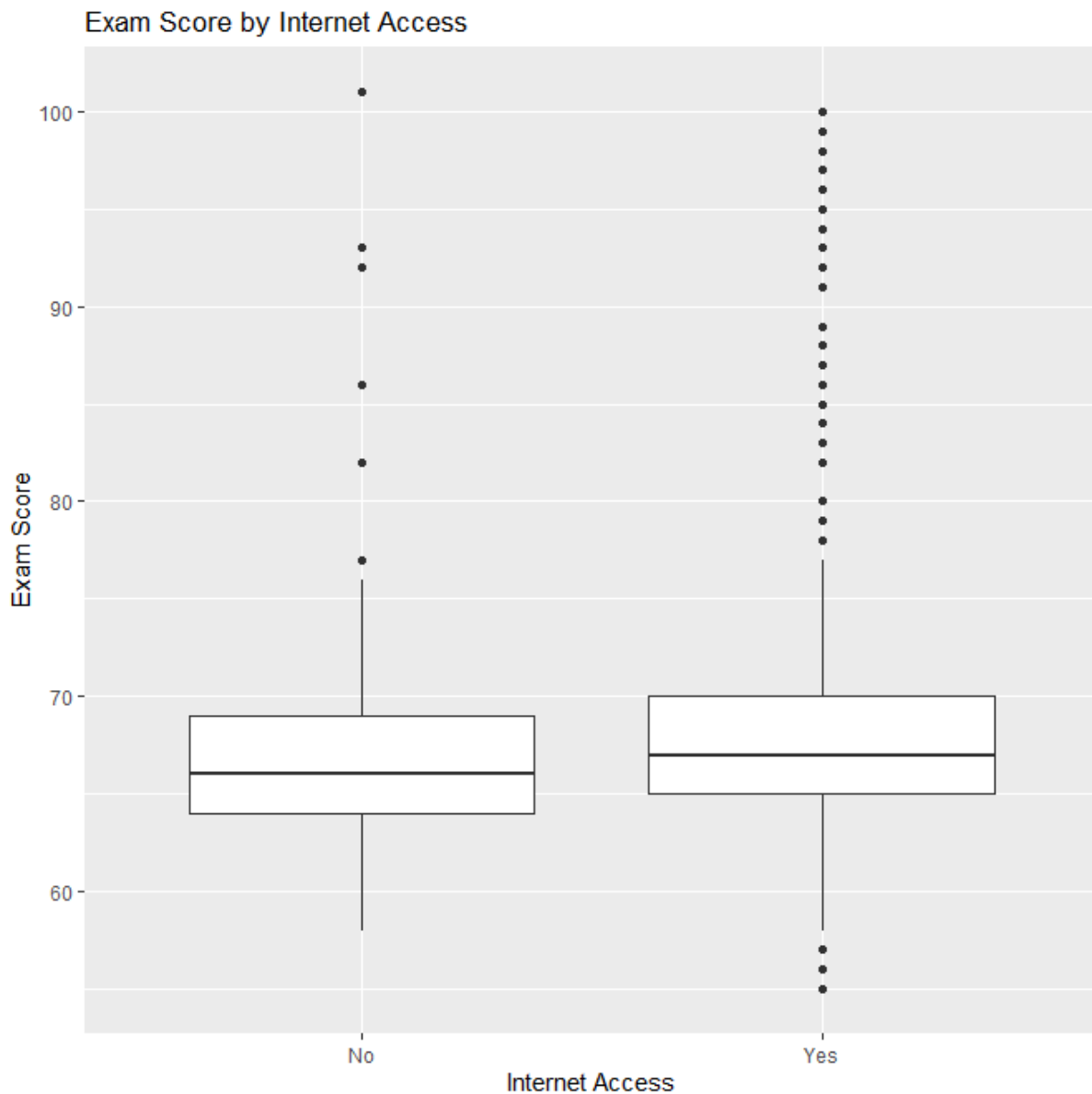


figure 5.2.1.1 Boxplot of Exam Score by Internet Access

The boxplot compares exam scores between students with and without Internet access, allowing for an easy visual comparison of their performance. Overall, students with Internet access show a slightly higher median exam score, suggesting a modest advantage compared to those without access. Both groups display a similar interquartile range, indicating that the variability of exam performance is relatively consistent. Outliers are present in both categories, showing that some students perform much higher or lower than the typical range regardless of Internet access. Despite the overlap between the two distributions, the slight upward shift in the “Yes” category provides some support for Hypothesis 2 (students with Internet access tend to achieve better exam results).

6.2.2 Result and Conclusion

```
> df %>% group_by(internet_access) %>% summarize(n=n(), mean_score = mean(exam_score, na.rm=TRUE), sd=sd(exam_score, na.rm=TRUE))
# A tibble: 2 × 4
  internet_access     n mean_score     sd
  <fct>          <int>      <dbl> <dbl>
1 No             499       66.5  4.12
2 Yes            6108       67.3  3.87
> t_test_h2 <- t.test(exam_score ~ internet_access, data = df, alternative = "less", var.equal = FALSE)
> t_test_h2

Welch Two Sample t-test

data: exam_score by internet_access
t = -3.9689, df = 571.99, p-value = 4.07e-05
alternative hypothesis: true difference in means between group No and group Yes is less than 0
95 percent confidence interval:
 -Inf -0.443242
sample estimates:
mean in group No mean in group Yes
 66.53507         67.29289
```

figure 5.2.2.1 Two-sample t-test of Hypothesis 2

Two-sample t-test result:

- A Welch two-sample t-test was conducted to compare exam scores between students with and without internet access.
- Students with internet access had a higher mean exam score (67.29) compared to students without internet access (66.54).
- The test statistic was $t = -3.9689$, with $df = 571.99$, indicating a meaningful difference between the groups.
- The p-value was extremely small ($p = 4.07 \times 10^{-5}$), providing strong evidence to reject the null hypothesis.
- The one-sided confidence interval $(-\infty, -0.443)$ suggests that the true mean difference (No/Yes) is clearly less than zero.
- These results support the alternative hypothesis: students with internet access score significantly higher than those without internet access.

Formula:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Where:

- $\bar{x}_1$: mean of group 1 (No internet)
- $\bar{x}_2$: mean of group 2 (Yes internet)
- s_1^2 : variance of group 1

- s_2^2 :variance of group 2
- n_1 :sample size of group 1
- n_2 :sample size of group 2

Conclusion

The results of the two-sample t-test indicate that students with internet access score significantly higher on exams compared to those without access. The difference in mean exam scores is statistically significant, and the direction of the results supports the hypothesis that having internet access is associated with improved academic performance. Therefore, we conclude that internet access has a positive impact on students' exam outcomes in this dataset.

6.3 Statistics Testing 3: ANOVA

A single-way ANOVA was used to compare the means of more than two groups at once in order to see whether exam scores varied among the three levels of parental participation (Low, Medium, and High). A post-hoc Tukey HSD test was used to determine which particular groups differed from one another while accounting for Type I error because the ANOVA findings indicated a significant overall difference. This set of tests guarantees accurate pairwise comparisons as well as an overall evaluation of group differences.

6.3.1 Data Visualization 3: Boxplot of Hypothesis 3

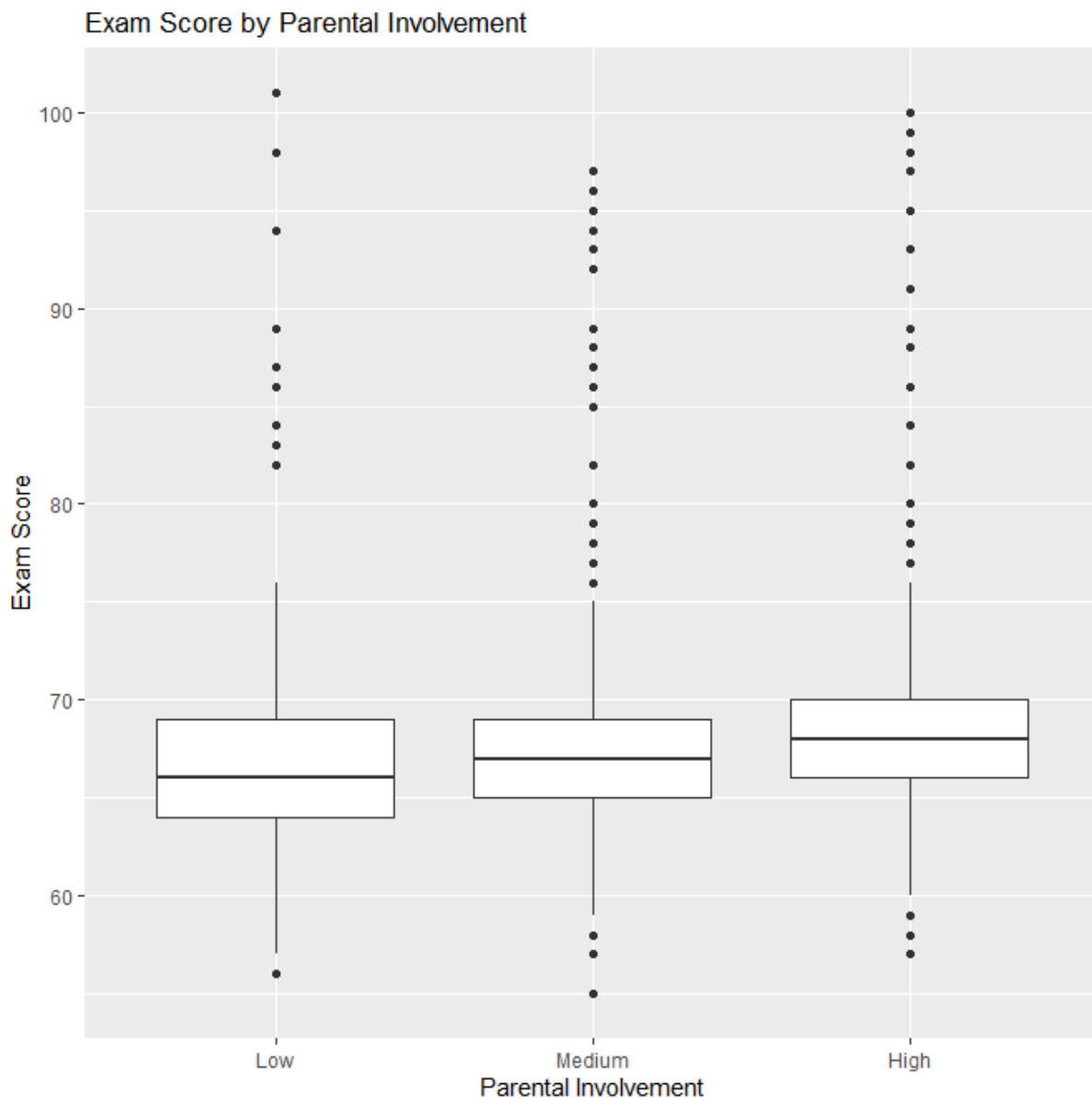


figure 6.3.1.1 Boxplot of Exam Score by Parental Involvement

The boxplot compares exam scores across three levels of parental involvement which include low, medium, and high to examine whether greater involvement is associated with better academic performance.

Overall, the median exam scores show a slight upward trend from low to high involvement, indicating that students with more supportive parental engagement tend to achieve marginally higher scores. The interquartile ranges across all three groups are similar, suggesting comparable variability in performance regardless of involvement level. Outliers are present in each category, showing that individual performance can vary widely even within the same involvement group. Although the differences between the groups are not substantial, the

gradual increase in median values provides some support for Hypothesis 3 (higher parental involvement is linked to improved exam outcomes).

6.3.2 Result and Conclusion

```
> aov_h3 <- aov(exam_score ~ parental_involvement, data = df)
> summary(aov_h3)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
parental_involvement	2	2494	1247.2	84.49	<2e-16 ***
Residuals	6604	97492	14.8		

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> # Post-hoc if ANOVA significant
> TukeyHSD(aov_h3)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = exam_score ~ parental_involvement, data = df)

$parental_involvement
      diff      lwr      upr p adj
Medium-Low  0.7398911 0.4486729 1.031109    0
High-Low    1.7345025 1.4132602 2.055745    0
High-Medium 0.9946114 0.7364463 1.252777    0
```

figure 6.3.2.1 ANOVA Testing of Hypothesis 3

ANOVA Result:

- **F value: 84.49**
- **p-value: < 2e-16**
- There is a significant difference in exam scores between at least two parental involvement groups.
- ANOVA tells us a difference exists, but not which groups are different.

Tukey HSD:

1. Medium vs Low

- diff = 0.74
- Meaning: Medium involvement students score 0.74 points higher than Low
- lwr and upr CI are positive which is significant
- p adj = 0 which is very strong significance

2. High vs Low

- diff = 1.73
- High involvement students score 1.73 points higher than Low

- Confidence interval does not include 0 which is significant
- $p_{adj} = 0$

3. High vs Medium

- $diff = 0.99$
- High involvement students score 0.99 points higher than Medium
- Significant difference
- $p_{adj} = 0$

Conclusion:

- ANOVA found a significant effect of parental involvement on exam scores, $F(2, 6604) = 84.49, p < 2e-16$.
- Students with medium involvement scored about 0.74 points higher than those with low involvement.
- Students with high involvement scored about 1.73 points higher than those with low involvement.
- Students with high involvement also scored 0.99 points higher than those with medium involvement.
- Overall, exam scores increase consistently from low to medium to high parental involvement.

7.0 Regression Analysis

Regression analysis is used to examine how one or more independent variables predict a continuous outcome. In this chapter, regression helps determine how different factors influence exam performance and how strongly each variable contributes to the results. This method allows us to understand relationships within the data and evaluate how well the model explains student outcomes.

7.1 Multiple Linear Regression

Linear Regression is a statistical technique used to analyze the relationship between two variables, helping researchers determine the strength and direction of their association. When a meaningful relationship is found, the model can be used to predict the dependent variable from the independent variable. When multiple predictors are involved, the method becomes Multiple Linear Regression (MLR), which assesses how several factors together influence the outcome. [9]

The general form of a Multiple Linear Regression model is expressed as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \varepsilon$$

Where:

- **Y:** Dependent (output) variable
- $X_1, X_2, \dots, X_n$: Independent (input) variables
- β_0 : Intercept
- $\beta_1, \beta_2, \dots, \beta_n$: Regression coefficients that represent the influence of each predictor
- ε : Erro term

In this study, the dependent variable is Exam Score, and the independent variables include factors that may influence academic performance. Thus, the Multiple Linear Regression model for this analysis can be represented as:

$$\begin{aligned} \text{Exam Score} = & \beta_0 + \beta_1(\text{Hours Studied}) + \beta_2(\text{Sleep Hours}) + \beta_3(\text{Motivation Level}) \\ & + \beta_4(\text{Internet}) + \varepsilon \end{aligned}$$

Where:

- **Hours Studied** represents how much time the student studies daily
- **Sleep Hours** reflects the amount of rest a student gets each day
- **Motivation Level** is an ordered factor (Low, Medium, High) indicating academic motivation
- **Internet Access** (Yes/No) serves as a proxy for potential online distraction or support
- **Exam Score** is the predicted academic performance

7.1.1 Data Visualization 4: Regression Diagnostic Plot Interpretation

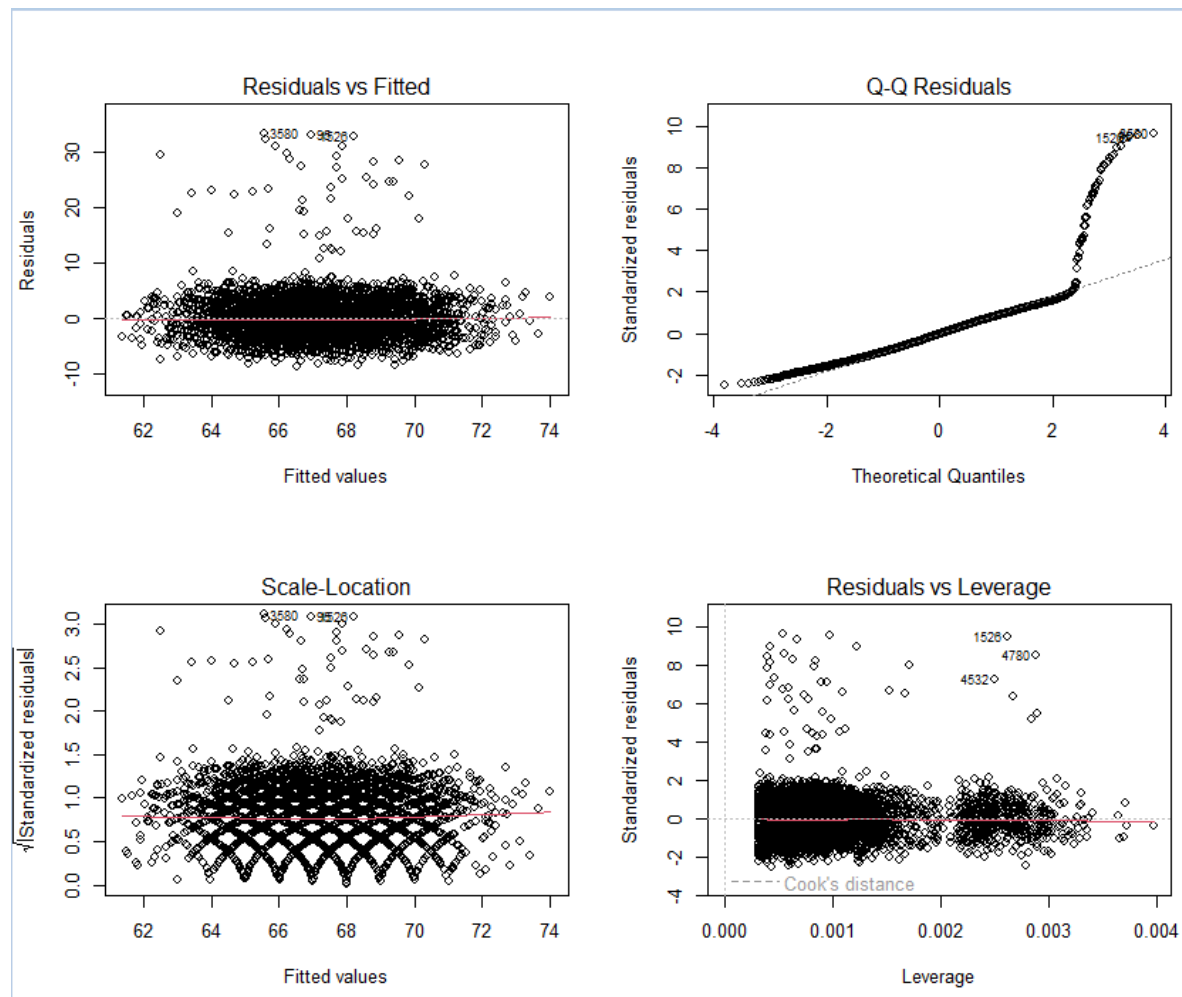


figure 7.1.1.1 Regression Diagnostic Plot Interpretation

1. Residuals vs Fitted

- Residuals are generally centered around zero, which supports the assumption of linearity between the predictors and the outcome.
- The plot does not show a strong curved pattern, indicating that the model form is mostly appropriate.
- However, a slight widening of the residual spread is visible as fitted values increase, suggesting mild heteroscedasticity (unequal variance).
- A few residuals appear much higher or lower than the rest, indicating possible outliers.

2. Q-Q Residuals

- Residuals follow the diagonal line fairly well in the middle range, showing that the central portion of the data is approximately normally distributed.

- The upper tail deviates significantly upward, indicating several extreme positive residuals (over-predictions or unusual high errors).
- This deviation suggests that the assumption of normality of residuals is not fully met.
- The presence of these deviations may affect statistical tests such as p-values and confidence intervals.

3. Scale–Location Plot

- The plot shows the square-root standardized residuals against fitted values to examine variance consistency.
- The spread across the fitted values is not uniform; instead, a slightly increasing trend appears.
- This pattern confirms that variance is not constant, indicating heteroscedasticity.
- Heteroscedasticity can reduce the reliability of standard errors and significance tests.

4. Residuals vs Leverage

- Most observations fall within Cook's distance threshold, meaning they do not exert excessive influence on the regression model.
- Some of these high-leverage points also have larger residuals, making them potentially influential and capable of distorting regression coefficients.
- These points may require further investigation or sensitivity testing to ensure they do not disproportionately affect the model.

7.1.2 Result and Conclusion

```
> lm_full <- lm(exam_score ~ hours_studied + sleep_hours + motivation_level + internet_access, data = df)
> summary(lm_full)

Call:
lm(formula = exam_score ~ hours_studied + sleep_hours + motivation_level +
    internet_access, data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-8.550 -2.252 -0.143  1.978 33.439

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   61.258602   0.290414  210.935 < 2e-16 ***
hours_studied    0.289858   0.007109   40.775 < 2e-16 ***
sleep_hours     -0.059849   0.029001   -2.064  0.0391 *
motivation_level.L  0.712851   0.087365    8.159 4.00e-16 ***
motivation_level.Q -0.024870   0.070192   -0.354  0.7231
internet_accessYes  0.702655   0.161145    4.360 1.32e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.46 on 6601 degrees of freedom
Multiple R-squared:  0.2096,    Adjusted R-squared:  0.209
F-statistic: 350.1 on 5 and 6601 DF,  p-value: < 2.2e-16

> confint(lm_full)
              2.5 %      97.5 %
(Intercept)  60.6892957  61.827907477
hours_studied    0.2759224  0.303793118
sleep_hours     -0.1167003 -0.002996875
motivation_level.L  0.5415871  0.884115212
motivation_level.Q -0.1624700  0.112729059
internet_accessYes  0.3867574  1.018551855
```

figure 7.1.2.1 Multiple Linear Regression Testing of Hypothesis 4

1. Residuals:

- Min = -8.55
- Max = 33.44
- These indicate reasonable variation, with no severe outliers indicated by the summary.

2. Multiple Linear Regression Coefficients

Table 7.1.2.2 Multiple Linear Regression Coefficients

Predictor	Estimate	Meaning
Intercept	61.2586	Predicted exam score when all predictors = 0 (baseline)
Hours_studied	0.289858	Each additional hour studied increases exam score by 0.29 points (significant)
Sleep_hours	-0.059849	Each extra hour of sleep slightly reduces score by 0.06 points (small but significant)
Motivation_level.L (Low)	0.712851	Low motivation students score 0.71 points more than the baseline group (depends on factor coding)
Motivation_level.Q (Medium)	-0.052708	Not significant (p = 0.7231)
Internet_accessYes	0.702655	Students with internet access score 0.70 points higher on average

Summary of Multiple Linear Regression Coefficients:

A. Significant predictors:

- Hours_studied (very strong)
- Sleep_hours (small effect but significant)
- Motivation_levelL
- Internet_accessYes

B. Not significant:

- Motivation_levelQ (no meaningful difference compared to baseline)

3. Model Fit Statistics

Table 7.1.2.3 Model Fit Statistics

Statistic	Value	Meaning
Residual Standard Error	3.46	Average difference between predicted and actual exam scores.
Degrees of Freedom	6601	Remaining variance after accounting for predictors.
Multiple R-squared	0.2096	The model explains 20.96% of variance in exam scores.
Adjusted R-squared	0.209	R ² adjusted for number of predictors; similar value indicates no overfitting.
F-statistic	350.1	Tests whether the overall model is significant.
F-statistic df	5, 6601	5 predictors, 6601 residual df.
Model p-value	< 2.2e-16	The full regression model is statistically significant.

4. Confidence Intervals

Table 7.1.2.4 Confidence Intervals

Predictor	Lower Bound (2.5%)	Upper Bound (97.5%)	Interpretation
(Intercept)	60.6893	61.8279	The baseline predicted exam score (when all predictors = 0) lies between 60.69 and 61.83.
Hours_studied	0.2736	0.3061	Each additional study hour increases exam score by 0.27 to 0.31 points; consistently positive.
Sleep_hours	-0.1167	-0.0030	Sleep hours have a small negative effect (-0.12 to -0.003); interval does not include zero → significant.
Motivation_level.L (Low)	0.5488	0.8769	Students with motivation level “L” score 0.55 to 0.88 points higher than the baseline group.

Motivation_level.Q (Medium)	-0.4674	0.3621	Interval includes zero → no significant difference between this group and baseline.
Internet_accessYes	0.3864	1.0190	Students with internet access score 0.39 to 1.02 points higher than those without access.

8.0 Model Evaluation

Model evaluation determines how well a regression model fits the data and how accurately it predicts the outcome variable. After building the model, it is important to assess its performance using metrics such as R^2 , Adjusted R^2 , MSE, and RMSE, which indicate how much variance is explained and how large the prediction errors are. Diagnostic checks, including residual plots and normality tests, further ensure that the model meets the assumptions of linear regression. Together, these measures help verify whether the model is reliable and suitable for interpreting or predicting exam performance.

```
> model_glance <- broom::glance(lm_full)
> r_squared <- model_glance$r_squared
> adj_r_squared <- model_glance$adj.r_squared
>
> preds <- predict(lm_full, newdata = df)
> resid <- df$exam_score - preds
> mse <- mean(resid^2, na.rm = TRUE)
> rmse <- sqrt(mse)
>
> cat("R2 =", r_squared, "\nAdjusted R2 =", adj_r_squared, "\nMSE =", mse, "\nRMSE =", rmse, "\n")
R2 = 0.2095947
Adjusted R2 = 0.208996
MSE = 11.96148
RMSE = 3.458538
```

figure 8.0.1 R code of Model Evaluation

1. $R^2 = 0.2096$ ($\approx 21\%$)

- R^2 tells you how much of the variation in Exam Score is explained by the model.
- An R^2 of 0.21 means the predictors (Study Hours, Sleep Hours, Motivation Level, Internet Access) explain 21% of the differences in students' exam scores.
- This indicates a weak-to-moderate model fit.
- The remaining 79% of variation is explained by other factors not included in the model (e.g., teaching quality, prior knowledge, stress levels, socioeconomic factors).

2. Adjusted $R^2 = 0.2090$

- Adjusted R^2 is a more accurate version of R^2 when there are multiple predictors.
- Since Adjusted R^2 (0.2090) is very close to R^2 (0.2096)
- This means the predictors added to the model are relevant and not unnecessarily inflating R^2 .

The model is not overfitted.

3. MSE = 11.96

- MSE (Mean Squared Error) measures the average squared distance between predicted and actual exam scores.
 - A lower MSE indicates better accuracy.
 - An MSE of 11.96 means the model has some prediction error but is reasonably stable for educational data.
4. RMSE = 3.46
- RMSE is the square root of MSE, and it is easier to interpret because it is in the same units as the exam score.
 - This means the model's predictions are typically off by about 3–4 exam score points.
 - In education datasets, this is considered acceptable but not highly accurate.

9.0 Methodology

This study analyzes the factors affecting the academic success of students using a quantitative, statistical research methodology. Numerical analysis of data, hypothesis testing, and the use of statistical techniques to determine correlations between variables are all suitable uses for quantitative methods [10]. Through statistical testing and predictive modeling, the methodology is intended to methodically examine the connections between students' exam performance and internal, external, lifestyle, and technology use factors. Regression modeling, hypothesis testing, exploratory analysis, dataset selection, data preprocessing, and model evaluation contribute to the whole research process.

9.1 Research Design

This study uses a cross-sectional quantitative research design. This research analyzes secondary data from an open-source dataset to find patterns, relationships, and predictive factors affecting student performance. Cross-sectional designs are important in behavioral and educational research because they allow statistical inference without requiring long-term data collection and allow effective analysis of huge datasets [11]. This methodology works since the study is focused more with statistical analysis and numerical measures rather than qualitative interpretation.

9.2 Dataset and Data Collection

The dataset used in this research is the Student Performance Factors dataset obtained from Kaggle, consisting of 6,607 student records and 20 variables. One variable, ExamScore, serves as the dependent variable, while selected independent variables represent study habits, motivation, parental involvement, lifestyle factors, and technology usage. The dataset includes a mixture of continuous, categorical, and ordinal data, making it appropriate for statistical testing and regression modeling commonly used in educational data analysis [12].

9.3 Variables Selection

Key independent variables like study hours, sleep hours, motivation level, involvement of parents, internet access, physical activity, and tutoring sessions were selected based on their relevance to the research objectives. These variables show that academic achievement is influenced by both internal and external factors. Students' academic performance is measured on a continuous scale by the dependent variable, ExamScore, which is appropriate for regression and correlation analysis. [13].

9.4 Data Preprocessing and Cleaning

The collection of data was performed to ensure accuracy and dependability during analysis. Inconsistencies and missing values were examined for in the dataset. The lack of missing data in the results showed great data quality. In order to simplify statistical testing and regression analysis, categorical variables were correctly encoded. Graphical and statistical methods were used to look for outliers in continuous data, and when necessary, assumptions of normality and linearity were evaluated to ensure reliable inferential conclusions [12].

9.5 Statistical Analysis Techniques

Several statistical methods were applied based on variable types and research hypotheses

Table 9.5.1 Statistical Testing Techniques

Statistical Testing Techniques	Description
Pearson Correlation Analysis	Used to examine the strength and direction of linear relationships between continuous variables, such as study hours and exam scores.
Two-sample (Welch) t-tests	Conducted to compare mean exam scores between two independent groups, particularly students with and without internet access. This test is robust when group variances are unequal.
One-way ANOVA	Applied to compare exam scores across more than two groups, such as levels of parental involvement, followed by Tukey HSD post-hoc tests to identify specific group differences while controlling for Type I error.

9.6 Multiple Linear Regression Modeling

A Multiple Linear Regression (MLR) model was developed to predict students' exam scores using multiple independent variables simultaneously. The regression model evaluates the

combined effect of study hours, sleep hours, motivation level, and internet access on academic performance. Multiple linear regression is widely used in educational research to assess the relative contribution of multiple predictors to a continuous outcome variable [14]. Model assumptions, including linearity, normality of residuals, homoscedasticity, and influential observations, were assessed using regression diagnostic plots.

9.7 Model Validation and Evaluation

Model performance was assessed using R^2 , Adjusted R^2 , Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). These evaluation metrics are commonly used to measure model fit, explanatory power, and prediction accuracy in regression-based studies [15]. Diagnostic checks were also performed to ensure that regression assumptions were reasonably met.

10.0 Results and Discussion

10.1 Results

Statistical analysis showed that several factors significantly affect student academic performance. Pearson correlation revealed a moderate positive relationship between study hours and exam scores, indicating that students who study more tend to achieve higher results. The two-sample t-test showed that students with internet access scored significantly higher than those without access. One-way ANOVA results also confirmed significant differences in exam scores across levels of parental involvement, with higher involvement associated with better performance.

Multiple Linear Regression analysis demonstrated that study hours, motivation level, sleep hours, and internet access collectively had a significant effect on exam scores. The model explained approximately 21% of the variance in academic performance, with study hours being the strongest positive predictor.

10.2 Discussion

The results indicate that academic performance is influenced by both individual behavior and external support factors. Study habits play a major role in determining exam outcomes, reinforcing the importance of effective time management. Internet access appears to support learning by providing educational resources, although its impact is relatively modest. Parental involvement also contributes positively, highlighting the importance of family support in students' academic success. The moderate model fit suggests that other factors not included in the study may also influence performance.

11.0 Conclusion and Future Work

11.1 Conclusion

In conclusion, this study confirms that study hours, parental involvement, motivation, and internet access significantly influence student academic performance. Study habits emerged

as the most influential factor. Although the regression model does not explain all variations in exam scores, it provides useful insights for developing targeted educational strategies. Future research should include additional factors and advanced modeling techniques to improve predictive accuracy.

11.2 Limitation

- The dataset is cross-sectional, limiting causal interpretation.
- The regression model explains only part of the variance in exam scores, indicating that important factors such as teaching quality, mental health, and socioeconomic status were not included.
- Some variables, such as motivation and parental involvement, rely on categorical or self-reported measures, which may introduce subjectivity.

11.3 Future Work

- Incorporating additional variables such as teacher quality, learning styles, stress levels, and socioeconomic background to improve model accuracy.
- Applying advanced machine learning techniques (e.g., random forest, gradient boosting) to capture nonlinear relationships.
- Conducting longitudinal studies to examine how academic performance evolves over time.
- Exploring interaction effects, such as how internet usage moderates the relationship between study habits and exam performance.

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